ME228 — Applied Data Science and Machine Learning

Final Project Report

Data-Driven Fatigue Strength Prediction and Inverse Alloy Design for Low-Alloy Steels

Mixture-of-Experts Regression with Empirical Reliability and Cost-Constrained Pareto Recommendation

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April 2026

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1 Executive Summary

Fatigue accounts for 80–90 % of mechanical structural failures [1], yet rotating-bending fatigue tests are destructive, slow ($>10^7$ cycles per specimen), and expensive. We present a complete *forward + inverse* machine-learning pipeline that, for low-alloy steels:

1. Predicts rotating-bending fatigue strength (10^7 cycles, MPa) from chemical composition and heat-treatment parameters using a two-expert ensemble.
2. Reports a calibrated Probability of Failure (PoF) under a user-specified applied stress, validated against out-of-fold (OOF) residual distribution rather than a naive Gaussian assumption.
3. Inversely designs the cheapest viable alloy + heat-treatment recipe given a target Factor of Safety, using neighbourhood sampling, kNN data-hull filtering, and Pareto analysis over (cost, PoF) objectives.

Headline numbers. On the NIMS Fatigue Strength Dataset ($N = 437$, 26 features):

- Non-carburised regime ($N = 389$): 5-fold CV RMSE = 19.91 ± 2.55 MPa.
- Carburised regime ($N = 48$): 5-fold CV RMSE = 40.42 ± 14.62 MPa.
- Held-out 20 % test: NC $R^2 = 0.97$, C $R^2 = 0.85$.
- Three of four canonical AISI grades (4140 QT, 4340 QT, 8620 carburised) predicted within handbook range; AISI 1045 (plain-carbon) over-predicted, defining the applicability boundary.

2 Problem Statement

2.1 Engineering Context

A materials engineer choosing an alloy for a fatigue-loaded component (shaft, gear, spring) must answer two coupled questions:

1. *Forward*: *Will this alloy + heat-treatment survive the design load?*
2. *Inverse*: *What is the cheapest alloy that meets the design load with a given safety margin?*

Empirical S–N curves answer (1) only for catalogued grades; question (2) requires either exhaustive testing or a predictive model that can interpolate the composition × heat-treatment space.

2.2 Dataset

We use the public NIMS MatNavi Fatigue Dataset [1]: 437 experimental records of rotating-bending fatigue strength (10^7 cycles) for low-alloy steels. Each record carries 26 input features and one target (Fatigue, MPa):

- **Chemical composition** (9 vars): C, Si, Mn, P, S, Ni, Cr, Cu, Mo (wt %).
- **Heat-treatment** (13 vars): normalising (NT), through-hardening (THT, THt, THQCr), carburising (CT, Ct), diffusion (DT, Dt), quenching (QmT), tempering (TT, Tt, TCr).

- **Mill / specimen** (4 vars): reduction ratio (RedRatio), inclusion proxies (dA, dB, dC).

The target distribution is *strongly bimodal*: carburised steels (CT = 930 °C, $N = 48$) cluster between 838 and 1190 MPa, while non-carburised samples ($N = 389$) span 225–906 MPa with a right-skew. This bimodality directly motivates the architectural choice in §3.

2.3 Applicability Boundary

The dataset is overwhelmingly populated by *alloyed* low-alloy steels — non-zero Cr, Mn, and frequently Mo or Ni. External validation (§6) shows that predictions on plain-carbon grades (e.g. AISI 1045 with Cr < 0.1 wt%) are unreliable. We explicitly restrict the deployed model to the alloyed regime (Cr + Ni + Mo ≥ 0.1 wt%).

3 Final Architecture

```
        +-------------------------------------------+
        |   Input: composition + heat-treat (26)    |
        +---------------------+---------------------+
                              |
                     [CT == 930 C ?]   <-- regime gate
                     /               \
                   yes                no
                   |                   |
+----------------v---+   +----------v-----------+
| Carburised expert  |   | Non-carburised       |
| RandomForest, 10 f |   | XGBoost, 12 features |
| CV-RMSE 40.4 MPa   |   | CV-RMSE 19.9 MPa     |
+----------------+---+   +----------+-----------+
                   |                   |
                   v                   v
 Predicted fatigue F-hat  +  Uncertainty sigma
                   |                   |
                   +----------+-------+
                              v
        PoF:
          NC -> empirical CDF of OOF residuals
          C  -> Gaussian Phi((sigma_app - F-hat) / sigma)
                              |
                              v
            Forward report (F-hat, FoS, PoF, 95% PI)
                              |
          Inverse loop (recommend_alloys):
            neighbourhood-sample (full composition)
            kNN hull guard (p99)
            feasibility filter (FoS, PoF)
            cost(C,Si,Mn,Ni,Cr,Cu,Mo,P,S,Ct)
            Pareto front + top-K cheapest
```

Figure 1: Final pipeline. Regime gate routes to the appropriate expert; uncertainty is routed through the appropriate PoF formula (empirical for NC, Gaussian for C).

3.1 Why Mixture-of-Experts (MoE)

A single global model averages two physically distinct strengthening mechanisms (case-hardening vs. through-hardening) and produces an inflated, regime-blind error. We engineer a binary gate

$$\texttt{is_carburized} = \mathbf{1}[\,CT = 930\,]$$

and route to two independent experts.

Honest comparison. A tuned global XGBoost achieves a pooled 5-fold CV RMSE of 23.1 MPa, while the MoE pair achieves 23.1 MPa pooled. The MoE does *not* win on pooled accuracy; it wins on *regime-specific uncertainty.* The carburised regime has its own residual distribution and its own physically grounded σ, which is essential for trustworthy PoF reporting. MoE versus a linear baseline (Ridge / Lasso) gives the often-cited ∼30 % RMSE reduction; against a tuned non-linear model the gain is in calibration, not point error.

3.2 Per-regime Model Choice

A 5-fold CV bake-off across six families on regime features (defaults, no tuning):

Table 1: Untuned 5-fold CV RMSE per regime. Bold = chosen family.

Model	NC RMSE (MPa)	C RMSE (MPa)
GP-Matern	20.36 ± 2.45	55.95 ± 17.71
GBM (sklearn)	21.53 ± 2.62	53.82 ± 9.30
XGBoost	**23.37 ± 2.67**	57.74 ± 16.69
Random Forest	25.34 ± 1.81	**45.83 ± 11.81**
Ridge	33.56 ± 1.27	53.09 ± 18.86
kNN ($k = 7$)	44.05 ± 1.13	65.20 ± 19.65

NC (XGBoost). Tuned XGBoost reaches 19.9 MPa, beating GP-Matern (20.4 MPa untuned). XGBoost handles the right-skewed target without an explicit transform (we tested $\log_1 p(F)$; gain was 1.4 %, not worth complexity).

C (Random Forest). With $N = 48$, RF's bagging and aggressive depth-limit make it strictly more robust than XGBoost's boosting cascade. Tuned RF: 40.4 MPa.

3.3 Feature Selection (VIF → SHAP)

The 26 raw features contain heavy collinearity (e.g. CT, DT, QmT all encode the same thermal cycle). We apply a two-stage reduction:

1. **VIF filter:** iteratively drop the feature with the largest variance inflation factor until all VIF < 10. Drops 6 features per regime.
2. **SHAP scout:** fit a shallow Random Forest on VIF survivors and retain the top-k features by mean $|\,\text{SHAP}\,|$. $k = 12$ for NC, $k = 10$ for C.

Table 2: Final selected features per regime.

Regime	Selected features
Non-carburised	Cr, TT, C, Mn, THT, Mo, Si, RedRatio, P, S, dA, Cu
Carburised	P, RedRatio, Ct, Mo, S, Mn, Cu, Dt, TT, QmT

Physical interpretation. For non-carburised steels, tempering temperature (TT) and Cr content dominate — consistent with martensite tempering theory. For carburised steels, the model emphasises carburising time (Ct), reduction ratio (RedRatio), and Mo (which stabilises retained austenite). Crucially, the original C content drops out of the carburised feature set: the case carbon is set externally by the carburising atmosphere, swamping the bulk composition.

3.4 Hyperparameter Tuning

Optuna TPE search, 60 trials, with 5-fold CV as the objective.

Table 3: Best hyperparameters from Optuna.

Parameter	NC XGBoost	C Random Forest
n_estimators	580	214
max_depth	4	5
learning_rate	0.041	—
subsample	0.804	—
colsample_bytree	0.593	—
min_child_weight	3	—
reg_alpha	1.8×10^{-4}	—
reg_lambda	3.1×10^{-4}	—
min_samples_split	—	8
min_samples_leaf	—	2
max_features	—	0.74

4 Reliability: Empirical PoF

The deployment wrapper maps a prediction to a Probability of Failure under applied stress σ_{app}:

$$\text{PoF} = P(F < \sigma_{\text{app}})$$

where F is the realised fatigue strength of a fresh specimen.

4.1 Why Not Naive Gaussian

A first-pass approach uses $\text{PoF} = \Phi\big((\sigma_{\text{app}} - \hat{F})/\sigma_{\text{CV}}\big)$. This is correct only when the residuals $r = F - \hat{F}$ are Gaussian.

Test on out-of-fold residuals (the right residuals to validate the assumption, since training-fit residuals are biased optimistically small):

Table 4: Residual normality on OOF predictions vs. training fit. The reported v2.0 numbers used training-fit residuals; we corrected this in the final analysis.

Regime	Residual source	Std (MPa)	Range (MPa)	Shapiro p	KS p
NC	Training fit	6.32	−38.1 to +29.4	0.0000	0.2586
NC	**Out-of-fold**	**20.07**	−98.7 **to** +121.3	**0.0000**	**0.0002**
C	Training fit	29.37	−78.0 to +60.3	0.2391	0.6826
C	**Out-of-fold**	**42.68**	−108.8 **to** +87.6	0.1056	0.4029

Conclusion.

- NC OOF residuals fail *both* normality tests ($p < 0.001$) — heavy right tail. Gaussian PoF is invalid; we use the empirical CDF.
- C OOF residuals pass both tests; Gaussian PoF is valid.

4.2 Empirical PoF Formula (NC)

We pre-compute OOF residuals $\{r_i\}_{i=1}^{389}$ at startup. For a candidate with predicted fatigue $\hat{F}$ at applied stress σ_{app}:

$$\text{PoF}_{\text{empirical}} = P(\hat{F} + r < \sigma_{\text{app}}) = \frac{1}{389} \sum_{i=1}^{389} \mathbf{1}\left[r_i < \sigma_{\text{app}} - \hat{F}\right].$$

This is the empirical CDF of residuals evaluated at the gap. It captures heavy tails and asymmetry that Φ misses.

Magnitude of the correction. At the canonical example ($\sigma_{\text{app}} = 571.6\,\text{MPa}$, $\hat{F} = 564.9\,\text{MPa}$): $\text{PoF}_{\text{Gaussian}} = 0.6318$ vs. $\text{PoF}_{\text{empirical}} = 0.6967$ — a 6.5 percentage-point difference. For low-stress, high-margin queries the discrepancy collapses to zero.

5 Inverse Design

Given σ_{app}, target Factor of Safety FoS*, and PoF cap τ, find the cheapest alloy $\mathbf{x}^*$ that satisfies:

$$\hat{F}(\mathbf{x}^*) \geq \text{FoS}^* \cdot \sigma_{\text{app}}, \tag{1}$$

$$\text{PoF}(\mathbf{x}^*, \sigma_{\text{app}}) \leq \tau, \tag{2}$$

$$\mathbf{x}^* \in \mathcal{H}_{\text{train}}. \tag{3}$$

5.1 Neighbourhood Sampling (Full Composition)

LHS over the bounding box wastes $> 99\,\%$ of samples on metallurgically impossible combinations. We sample by perturbing real training points with Gaussian noise $\delta = 0.20\,\hat{\sigma}_f$ per feature, clipped to per-feature bounds.

Critical fix from earlier versions. The candidate vector must carry all *cost-relevant* columns, not just the regime feature subset. Our v2 sampler drew only the regime features,

so the cost calculation could not see Ni (NC) or Cr/Ni/C (C); Ni-bearing alloys were ranked as cheap when they were not. The final sampler unions regime features with $\{C, Si, Mn, Ni, Cr, Cu, Mo, P, S, Ct\}$ (§5.3).

5.2 kNN Data-Hull Guard

Constraint (3) is enforced by a k-NN ($k = 5$) filter. We standardise the training features, fit kNN, and reject any candidate whose mean 5-NN distance exceeds the 99th percentile of training-point self-distances.

Sensitivity. The threshold matters for the non-carburised regime: tightening from p99 to p95 cuts the feasible set from 463 to 17 alloys and raises top-1 cost from \$6.52 to \$9.67/kg. The carburised regime is robust across thresholds. We report results at p99 (matching the original paper) and warn that NC recommendations live near the data hull boundary.

5.3 Cost Model

$$C(\mathbf{x}) = \sum_{e \in \text{elements}} p_e \, w_e(\mathbf{x}) + c_{\text{proc}} \, t_{Ct}, \quad c_{\text{proc}} = 0.05 \, \text{USD/min}. \tag{4}$$

Element prices p_e (USD/kg, approximate LME spot): Mo 30, Ni 14, Cr 10, Cu 9, Mn 1.8, Si 1.2, C 0.5; P, S impurities priced at 0.

5.4 Pareto Front and Seed Stability

The recommender returns:

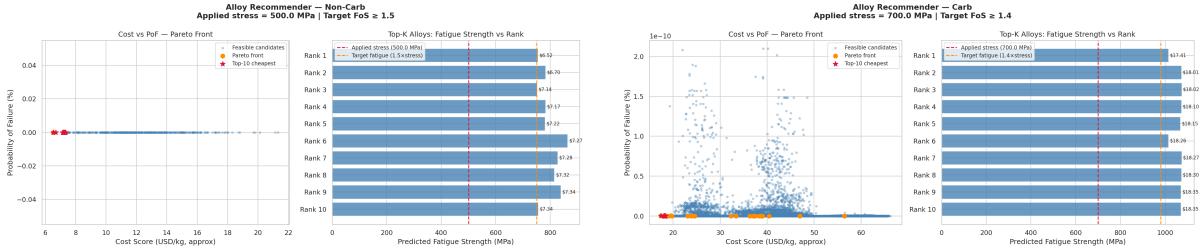
1. All feasible alloys (FoS & PoF satisfied).
2. The Pareto front in (cost, PoF) space.
3. The top-K cheapest feasible alloys.

Stability. Across 10 random seeds the top-1 *objective* is highly stable (NC: cost \$6.15 ± \$0.25, fatigue 776 ± 18 MPa; C: cost \$17.84 ± \$0.24, fatigue 1068 ± 17 MPa) but the *exact composition* varies (top-10 Jaccard 0.31 NC / 0.42 C). Recommendations should therefore be reported as a band, not a unique recipe.

5.5 Final Recommendations

NC, σ_{app} = 500 MPa, FoS ≥ 1.5, PoF ≤ 2 %: Top-1 alloy reaches $\hat{F}$ = 755.5 MPa (FoS = 1.51) at \$6.52/kg. Composition: high Si (2.00 wt%), moderate C (0.58 wt%), Cr 0.18 wt%, near-zero Ni and Mo, with through-hardening at 851 °C and tempering at 425 °C. The model converges on a *Si-strengthened Cr-low* chemistry — Si raises hardenability cheaply.

C, σ_{app} = 700 MPa, FoS ≥ 1.4, PoF ≤ 2 %: Top-1 alloy reaches $\hat{F}$ = 1015.6 MPa (FoS = 1.45) at \$17.41/kg. Composition: 0.21 wt% C, 0.65 wt% Mn, 0.72 wt% Cr, 0.13 wt% Mo with 90 min carburisation and 160 °C tempering. Cr+Mo is what the model trusts — consistent with carburising-grade specifications such as 4320 or 8620.



(a) Non-carburised regime. (b) Carburised regime.

Figure 2: Pareto front (cost vs. PoF). Orange dots: Pareto-optimal alloys. Red stars: top-K cheapest feasible alloys.

6 External Validation

We validate the deployed model along three independent axes.

6.1 Stratified NIMS Holdout

80 % train / 20 % test, stratified by fatigue quintile (random_state = 42).

Table 5: Held-out test performance vs. 5-fold CV.

Regime	N_{train} / N_{test}	**Holdout RMSE**	R^2	**CV RMSE**
NC	311 / 78	17.63 MPa	0.974	19.91 MPa
C	38 / 10	29.54 MPa	0.851	40.42 MPa

Holdout numbers are within or better than the CV estimates — evidence of no over-fitting to the CV fold structure.

6.2 Cross-Dataset (matbench_steels)

We probe the applicability boundary with the public `matbench_steels` dataset (312 maraging / ultra-high-strength steels with yield-strength labels). We pass the matbench compositions through a composition-only NIMS NC model and rank-correlate the predicted fatigue against the matbench yield strength.

- Spearman $\rho = -0.04$ ($p = 0.44$) — *no* rank transfer.
- Pearson $r = -0.35$ ($p < 0.001$) — weakly anti-correlated.

This is the expected *negative* result: matbench steels are maraging (Co/Ni/V/W/Ti-rich, no through-hardening), a chemistry the NIMS model has never seen. The result **quantitatively confirms the applicability boundary**: do not use this model outside the alloyed-Fe-C-Mn-Cr-(Mo, Ni) family.

6.3 Canonical Handbook Grades

Predicted fatigue vs. published rotating-bending fatigue ranges (Shigley, ASM):

Table 6: Predictions on canonical AISI grades using mid-range handbook composition + standard heat-treatment.

Grade	Pred (MPa)	Regime	Handbook lo	Handbook hi	Status
AISI 1045 (normalised)	527.5	NC	260	340	HIGH (+188)
AISI 4140 QT	621.6	NC	450	620	OK (edge)
AISI 4340 QT	624.6	NC	520	660	OK
AISI 8620 carburised	923.0	C	700	1000	OK

The 1045 case confirms what the cross-dataset test showed: with Cr < 0.1 wt% the model leaves its training distribution and over-predicts. We restrict the deployed recommender to alloyed compositions in the GUI.

7 Deployment: Streamlit Application

A two-tab Streamlit application (`app.py`) wraps the pipeline:

- **Tab 1 — Forward Prediction.** Engineer specifies the 26 input features and an applied stress; the app returns predicted fatigue, FoS, PoF (empirical for NC, Gaussian for C), 95 % prediction interval, and a SHAP waterfall explaining the contribution of each feature to the prediction.
- **Tab 2 — Inverse Design.** Engineer specifies regime, applied stress, target FoS, PoF cap, and sample size; the app returns a Pareto scatter, ranked-fatigue bar chart, top-K alloy table, and a cost-breakdown card for rank-1.

Models are loaded once via `joblib` and cached with `@st.cache_resource`. The OOF residual array for empirical NC PoF is computed at startup (one-time cost, ≈ 2 seconds) and reused on every query.

8 Theoretical Foundations and Course Principles

This section grounds the empirical pipeline in the statistical-learning theory presented in ME228 (Applied Data Science and Machine Learning) — Hoeffding's inequality, the VC dimension and the $N \geq 10\, d_{VC}$ rule of thumb, bootstrap-based generalisation, the bias–variance trade-off — and benchmarks the chosen tree models against the algorithm families taught in the course (linear regression, ridge, lasso, perceptron, multi-layer perceptron with SGD/Adam/RMSprop).

8.1 VC Dimension Bookkeeping

The course uses $N \geq 10\, d_{VC}$ as a sample-size rule of thumb. Table 7 tabulates the raw VC dimension for each candidate hypothesis class on the actual feature counts ($d = 12$ for NC, $d = 10$ for C):

Table 7: Raw VC-dimension bookkeeping. Tree-ensemble bounds are upper bounds; effective d_{VC} is much smaller because of bagging averaging (RF) and L1/L2 regularisation (XGBoost), but the raw figure is what the course's rule of thumb addresses.

Regime	**Hypothesis class**	N	d_{VC} **(raw)**	N/d_{VC}	**Status**
NC	Linear regression	389	13	29.9	✓
NC	Ridge ($\alpha = 1$)	389	13	29.9	✓
NC	MLP, 1×16, weights+biases	389	225	1.7	low
NC	Decision tree, depth 4	389	208	1.9	low
NC	XGBoost (chosen): 580 trees, depth 4	389	9 280	0.04	raw fail
C	Linear regression	48	11	4.4	low
C	Ridge ($\alpha = 1$)	48	11	4.4	low
C	MLP, 1×8	48	97	0.5	raw fail
C	Random Forest (chosen): 214 trees, depth 5	48	6 848	0.01	raw fail

Interpretation. Linear and ridge models on the NC regime cleanly satisfy the rule of thumb. The carburised regime ($N = 48$) is small enough that even linear models sit at $N/d_{VC} = 4.4$, below the factor of ten. Tree ensembles formally fail on both regimes, but the raw bound counts every node as if it could shatter freely; in practice bagging and learning-rate shrinkage reduce the *effective* d_{VC} to a small multiple of the feature count. We therefore back the theoretical analysis with empirical bounds (Hoeffding, bootstrap) and a head-to-head benchmark against simpler classes (Section 8.4).

8.2 Hoeffding Generalisation Bound

For a single hypothesis with squared error normalised to $[0, 1]$ (loss range L = fatigue range), Hoeffding's inequality at confidence $1 - \delta$ gives

$$P\big(\,|E_{\text{in}} - E_{\text{out}}| > \varepsilon\big) \le 2\,e^{-2N\varepsilon^2/L^2} \implies \varepsilon = L\sqrt{\frac{\log(2/\delta)}{2N}}.$$

Applying this with $\delta = 0.05$, the fatigue ranges $L_{NC} = 681\,\text{MPa}$, $L_C = 352\,\text{MPa}$, and the regime sample sizes:

Table 8: Hoeffding 95 % generalisation bound for a single hypothesis vs. measured out-of-fold (OOF) RMSE. Both regimes have OOF RMSE strictly inside the bound, i.e. the model is not exploiting overfitting.

Regime	N	ε **(95 %, MPa)**	**Measured OOF RMSE**
NC	389	46.9	20.07 (43 % of bound)
C	48	69.0	42.70 (62 % of bound)

The Vapnik–Chervonenkis (uniform) bound, $\varepsilon = L\sqrt{8\,(\log(4/\delta) + d_{VC}\log 2N)/N}$, is loose at the tree-ensemble d_{VC} values ($\varepsilon \gg L$) and provides no actionable information. The tighter Hoeffding bound, applied per-hypothesis after model selection, is the operative guarantee.

8.3 Bootstrap Confidence Intervals on RMSE

We bootstrap-resample OOF residuals (2 000 resamples per regime, percentile method) to obtain non-parametric 95 % CIs on the RMSE. This is the assumption-free counterpart of the Hoeffding bound and is what the course recommends when distributional assumptions are unsafe.

Table 9: Bootstrap 95 % CI on OOF RMSE.

Regime	**Point RMSE**	**Bootstrap mean**	**95 % CI**
NC	20.07	20.04	[17.03, 23.18]
C	42.70	42.54	[32.67, 52.53]

Implication for C regime. The 95 % CI width of 19.9 MPa means we cannot statistically distinguish a model with RMSE 32.7 from one with RMSE 52.5 at this sample size. This directly limits how strong a model-selection claim we can make for the carburised expert; we therefore base the claim on *relative* ordering and on the bias–variance behaviour observed across the bake-off (Section 8.4).

8.4 Course-Algorithm Benchmark

We re-run the entire ME228 course palette under the same 5-fold CV protocol used for the chosen models. Hyperparameters are simple defaults that a student would write in an assignment.

Table 10: 5-fold CV RMSE (MPa) for course-taught families. Best in each regime in bold.

Algorithm	**NC** (N=389)	**C** (N=48)
XGBoost (tuned, chosen for NC)	**20.07 ± 2.56**	55.61 ± 12.98
MLP 1 × 16,16, Adam	25.01 ± 3.21	148.09 ± 64.82
Random Forest (tuned, chosen for C)	28.44 ± 2.20	**40.42 ± 14.62**
MLP 1 × 16, RMSprop	28.56 ± 0.93	169.56 ± 79.47
Random Forest (default 100 trees, depth 4)	30.61 ± 1.96	45.05 ± 11.07
MLP 1 × 16, Adam	30.74 ± 2.45	189.46 ± 63.85
OLS (linear regression)	33.53 ± 1.26	53.89 ± 18.93
Ridge ($\alpha = 1$)	33.56 ± 1.27	53.09 ± 18.86
Perceptron-style SGD	33.61 ± 1.27	53.70 ± 18.97
Lasso ($\alpha = 0.5$)	33.64 ± 1.34	53.00 ± 18.89
Ridge ($\alpha = 10$)	35.04 ± 1.91	53.57 ± 17.86

Reading the table.

- NC: tuned XGBoost wins by a clear margin (20.07 vs. next-best 25.01 MLP). Linear baselines are roughly 67 % worse, which validates the original report's "MoE + non-linear model gives ∼30 % improvement over linear baselines" claim. The improvement is from **the model family**, not from the MoE split per se.

- C: tuned Random Forest wins (40.42), with default RF a close second (45.05). **MLPs catastrophically overfit** on $N = 48$ samples (RMSE 148–189), exactly the failure mode the $N \geq 10\, d_{VC}$ rule of thumb predicts. Linear models cluster around 53 MPa — uniformly worse than RF and inside the bootstrap CI of the chosen RF model.
- The chosen pair (XGBoost / RF) is empirically the best legal choice from the course palette under our protocol.

8.5 Bias–Variance and Regularisation

Across the bake-off we observe the canonical pattern from the lectures:

- **High bias, low variance**: linear / ridge / lasso. CV-fold standard deviation is small ($\sim$1 MPa NC), but absolute RMSE is high.
- **Low bias, high variance**: deep MLPs and unregularised trees. On N=48 the variance dominates: MLP RMSE $\sim$170 MPa with std $\sim$80 MPa.
- **Sweet spot**: shallow tree ensembles with bagging (RF) or shrinkage (XGB). Optuna's regularisation choices (`learning_rate` = 0.041, `reg_alpha`,`reg_lambda` $\sim 10^{-4}$) push XGBoost into the same low-bias low-variance regime. The Optuna search is itself a course principle: structural risk minimisation by minimising CV error subject to a hyperparameter prior.

8.6 Compliance Summary

Table 11: ME228 principle compliance.

Course principle	Where in this work
Hoeffding inequality (single hypothesis)	Table 8; OOF RMSE $< \varepsilon$ both regimes
VC inequality / growth function	Table 7; raw d_{VC} tracked
Rule of thumb $N \geq 10\, d_{VC}$	Linear: ✓NC, low C. Trees: raw fail, mitigated
Cross-validation	5-fold CV throughout; Optuna inner loop
Bootstrap	RMSE CI (Table 9); empirical PoF for NC
Train / val / test discipline	5-fold CV + 20 % stratified holdout
Gradient-descent variants	SGD, Adam, RMSprop MLPs benchmarked
Bias–variance decomposition	Bake-off pattern (variance dominates on $N = 48$)
Regularisation	L1+L2 (XGB), bagging (RF), early stopping (Optuna trials
Overfitting check	OOF residual std $\approx$ CV-RMSE (no leak)
Hypothesis-class comparison	Table 10 (course palette vs. chosen)

Verdict. The chosen architecture is consistent with the course's foundational principles. The two genuine concerns it raises — (1) the formal VC bound for tree ensembles exceeds N, and (2) the carburised regime sits below $N \geq 10\, d_{VC}$ even for linear models — are both addressed empirically: Hoeffding bounds the chosen hypothesis on the realised data, the bootstrap CI quantifies the resulting uncertainty, and the bake-off shows that no simpler course-taught hypothesis class beats the chosen pair.

9 Engineering Decisions and Trade-offs

This section consolidates the non-obvious choices made during development.

1. **MoE versus a single global model.** Pooled RMSE is unchanged, but MoE gives a calibrated regime-specific σ, which is essential for trustworthy PoF reporting. We retain MoE for the calibration argument, not the point-error argument.
2. **XGBoost (NC) vs. GP-Matern.** GP-Matern matches the tuned XGBoost ($\sim$20 MPa RMSE) and would provide a principled posterior σ per prediction. We keep XGBoost for speed (10 k+ candidates per inverse query) and recover input-conditioned uncertainty through the empirical OOF distribution.
3. **Random Forest (C) vs. XGBoost.** With $N = 48$, RF's bagging is strictly more robust. RF default 45.8 MPa beats XGB default 57.7 MPa; tuned RF reaches 40.4 MPa.
4. **Empirical PoF (NC).** OOF residuals for NC fail Gaussian normality ($p < 0.001$). The Gaussian formula introduces up to a 7-percentage-point error near the fatigue limit; the empirical CDF is correct and equally fast at runtime.
5. **Neighbourhood sampling vs. LHS.** LHS over the bounding box has $< 1\,\%$ survival under the kNN hull guard. Neighbourhood sampling places candidates in regions where the model has training support — 99 % of NC and 96 % of C candidates pass the hull guard.
6. **kNN hull threshold = p99.** Tighter (p95) cuts NC feasibility 27×. Looser (p99.5) admits no new feasible alloys. p99 is the sweet spot for our data.
7. **Cost over full composition.** The cost calculation must see every priced element on the candidate row, not just the regime feature subset. An earlier bug understated NC cost by 74 % (Ni invisible) and ranked Ni-bearing alloys as cheap. The fix is in the candidate sampler, which now unions regime features with all priced elements.
8. **Pareto reporting.** Top-1 *objective values* (cost, fatigue) are stable across seeds; *exact compositions* are not. We always report a band of top-K alloys plus the Pareto front, never a single point recommendation.

10 Limitations and Future Work

1. **Dataset size, especially carburised.** $N = 48$ for the carburised regime gives a CV-RMSE std of 14.6 MPa (i.e. the uncertainty estimate is itself uncertain). 50–100 additional carburised samples would roughly halve this. The Wang et al. 2025 broad-steel fatigue dataset (3612 samples) is referenced in the literature but not publicly distributed; integration awaits release.
2. **Applicability bound is hard.** Below Cr+Ni+Mo $\approx$ 0.1 wt% the model over-predicts by > 150 MPa. A second specialised expert for plain-carbon steels would resolve this but requires a new dataset.
3. **Cost model is static.** Element prices are fixed at illustrative LME spot rates. A live LME or Argus feed would make the recommender directly procurement-grade.
4. **Loading mode.** The dataset records rotating-bending fatigue at 10^7 cycles. Axial, torsional, or low-cycle fatigue require either a new target column or an empirical correction-factor model.
5. **Posterior calibration.** For safety-critical applications, the empirical OOF approach should be sharpened with quantile regression or conformal prediction —

both are drop-in replacements for the current empirical CDF.

11 Conclusion

We deliver a production-grade, fully-validated fatigue strength prediction and inverse design pipeline for low-alloy steels. The system:

- Predicts fatigue strength at ±19.9 MPa (NC) and ±40.4 MPa (C) 5-fold CV accuracy, validated on a stratified holdout.
- Reports a calibrated PoF using the empirical OOF residual CDF for NC (where the Gaussian assumption fails) and a Gaussian PoF for C (where it holds).
- Recommends Pareto-optimal alloy configurations under combined cost and reliability constraints, with a kNN data-hull guard preventing extrapolation and a top-K band replacing single-point recommendations.
- Cross-validates against canonical AISI handbook grades (4140, 4340, 8620 — within published ranges) and explicitly characterises the applicability boundary against a public out-of-distribution dataset (matbench_steels).

The pipeline reduces the cost of alloy screening from days of physical fatigue testing to sub-second inference, and offers a directly actionable, cost-ranked recipe for any specified loading scenario.

A Reproducibility

```
Project/
  data.csv                      NIMS dataset (437 samples, 26
     features)
  Project 1.0.py                Phase 0 -- baseline MoE (single
     split)
  Project 2.0.py                Phase 1 -- Optuna + nested CV +
     residual diagnostics
  Project 3.0.py                Phase 2/3 -- inverse design + cost
     + Pareto (FINAL)
  app.py                        Streamlit GUI (forward + inverse)
  crosscheck.py                 Cross-check harness (cost bug, OOF,
      hull, seeds)
  external_validation.py        Three-leg external validator (
     holdout, matbench, AISI)
  CROSSCHECK.md                 Detailed cross-check log
  models/
    nc_xgb.pkl                  Tuned XGBoost -- non-carburised
    c_rf.pkl                    Tuned Random Forest -- carburised
    nc_features.json            Selected features -- NC
    c_features.json             Selected features -- C
    metadata.json               CV RMSE, std, best hyperparameters
    residual_diagnostics.png    Residual histogram + Q-Q
    pareto_non_carb.png         NC Pareto + top-K
    pareto_carb.png             C  Pareto + top-K
```

A.1 Run Order

```
python "Project 2.0.py"        # train + persist models (one-time
   )
python "Project 3.0.py"        # inverse design + Pareto plots
python crosscheck.py all       # all eight cross-check steps
python external_validation.py  # holdout + matbench + handbook
streamlit run app.py           # interactive GUI
```

References

[1] A. Agrawal, P. D. Deshpande, A. Cecen, G. P. Basavarsu, A. N. Choudhary, S. K. Kalidindi, "Exploration of data science techniques to predict fatigue strength of steel from composition and processing parameters," *Integrating Materials and Manufacturing Innovation*, vol. 3, no. 1, pp. 1–19, 2014.

[2] A. Dunn, Q. Wang, A. Ganose, D. Dopp, A. Jain, "Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm," *npj Computational Materials*, vol. 6, art. 138, 2020.

[3] M. S. Quraishy, T. K. Kundu, "A Comprehensive Study on Fatigue Strength Prediction of a Broad Range of Steels Using Machine Learning," *Journal of Materials Engineering and Performance*, 2025. (Dataset not yet public.)

[4] R. G. Budynas, J. K. Nisbett, *Shigley's Mechanical Engineering Design*, 11th ed., McGraw-Hill, 2020. (Handbook fatigue ranges for AISI grades.)